Raed Najeeb

AI Engineer | Machine Learning | LLM Applications

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GitHub Portfolio

Professional Summary

AI Engineer and recent B.Tech graduate in Artificial Intelligence and Data Science with hands-on project experience building transformer models, retrieval-augmented generation (RAG) systems, and parameter-efficient fine-tuning workflows. Implemented a character-level transformer in PyTorch, developed a local RAG pipeline with vector retrieval and grounded generation, and evaluated LoRA-based emotion classification against full fine-tuning. Interested in AI Engineering, Machine Learning Engineering, and LLM application development. GATE DA qualified.

Technical Skills

Programming: Python
Machine Learning and Deep Learning: PyTorch, Transformers, Self-Attention, Embeddings, Model Training, Evaluation Metrics
LLM and Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, LoRA, QLoRA, Local LLM Inference
Libraries and Frameworks: Sentence-Transformers, ChromaDB, Hugging Face PEFT, Django, OpenCV, DeepFace
LLM Tools and Development: Ollama, Gemini API, Git, GitHub

Projects

nanoGPT From Scratch **2026**

- Implemented a character-level transformer in PyTorch, including tokenization, token embeddings, self-attention, causal masking, multi-head attention, feed-forward layers, residual connections, and the training loop.
- Debugged and corrected a causal masking error after identifying nonsensical text generation despite apparently strong training loss.
- Validated the corrected implementation using training and validation loss, reaching approximately 1.77 training loss and 1.90 validation loss.

RAG Pipeline **2026**

- Built a local retrieval-augmented generation pipeline with document ingestion, overlapping text chunking, Sentence-Transformers embeddings, ChromaDB vector storage, similarity-based retrieval, and context-grounded generation.
- Integrated a locally served Llama 3.1 8B model through Ollama to generate answers from retrieved document context.
- Achieved correct expected retrievals on all 8 labeled test questions and tested out-of-scope refusal behavior when the required information was unavailable.

Emotion Classification with LoRA **2026**

- Compared LoRA adaptation of DistilBERT with full fine-tuning on the six-class dair-ai/emotion dataset using held-out test evaluation.
- Achieved 91.65% accuracy and 0.877 macro F1 with the LoRA configuration, using approximately 1.10% of the model's parameters as trainable parameters.

CareerMate — Agentic AI Recruitment System **2025 – 2026**

- Contributed to a collaborative recruitment platform combining resume screening, aptitude evaluation, AI-assisted interviewing, behavioral analysis, and computer-vision proctoring.
- Worked with Django and the Gemini API for dynamic interview question generation, with OpenCV and DeepFace supporting computer-vision and behavioral-analysis features.
- Contributed to a recruiter dashboard aggregating candidate metrics and proctoring records.

Education

B.Tech in Artificial Intelligence & Data Science **Aug 2026**
Vimal Jyothi Engineering College, Kannur CGPA: 7.95 / 10

Achievements

GATE DA Qualified — 2025